Analytical Cumulative Examination
April 4, 1998

Recently, much interest has been expressed in a meteorite which, it has been suggested, came originally from Mars. Further, this interest has been heightened by the discovery of organized structures in the rock. These structures, it has been suggested, may be micro-fossils and thus may represent the remains of nano-bacteria. Alternately, it has been proposed that these entities represent natural, non-biogenic crystalline structures.

Several reports have recently appeared describing the characterization of this meteorite (Nature, Science, Chemical and Engineering News), and several presentations at the University of Arkansas have described, in detail the extensive analytical work performed on this sample (e.g. Professor Richard Zare, Stanford University).

Seven analytical methodologies are listed below which have been used to characterize this meteorite. Chose 5 of the 7 techniques, and for each choice provide a detailed overview of the analytical method including the basis for the analysis, and the type of information available. With respect to this particular meteoritic sample, describe what key information about the sample was obtained from the method (e.g. age, age on earth, time in space, origin: Mars, other, etc.).

Scanning Electron Microscopy

Laser-Desorption, Time-of-Flight Mass Spectrometry

Fluorescence Line Narrowing Spectroscopy

Electron Dispersive X-ray Fluorescence

C^{14} Analysis

Thermal Luminescence

Isotope Ratio Mass Spectrometry